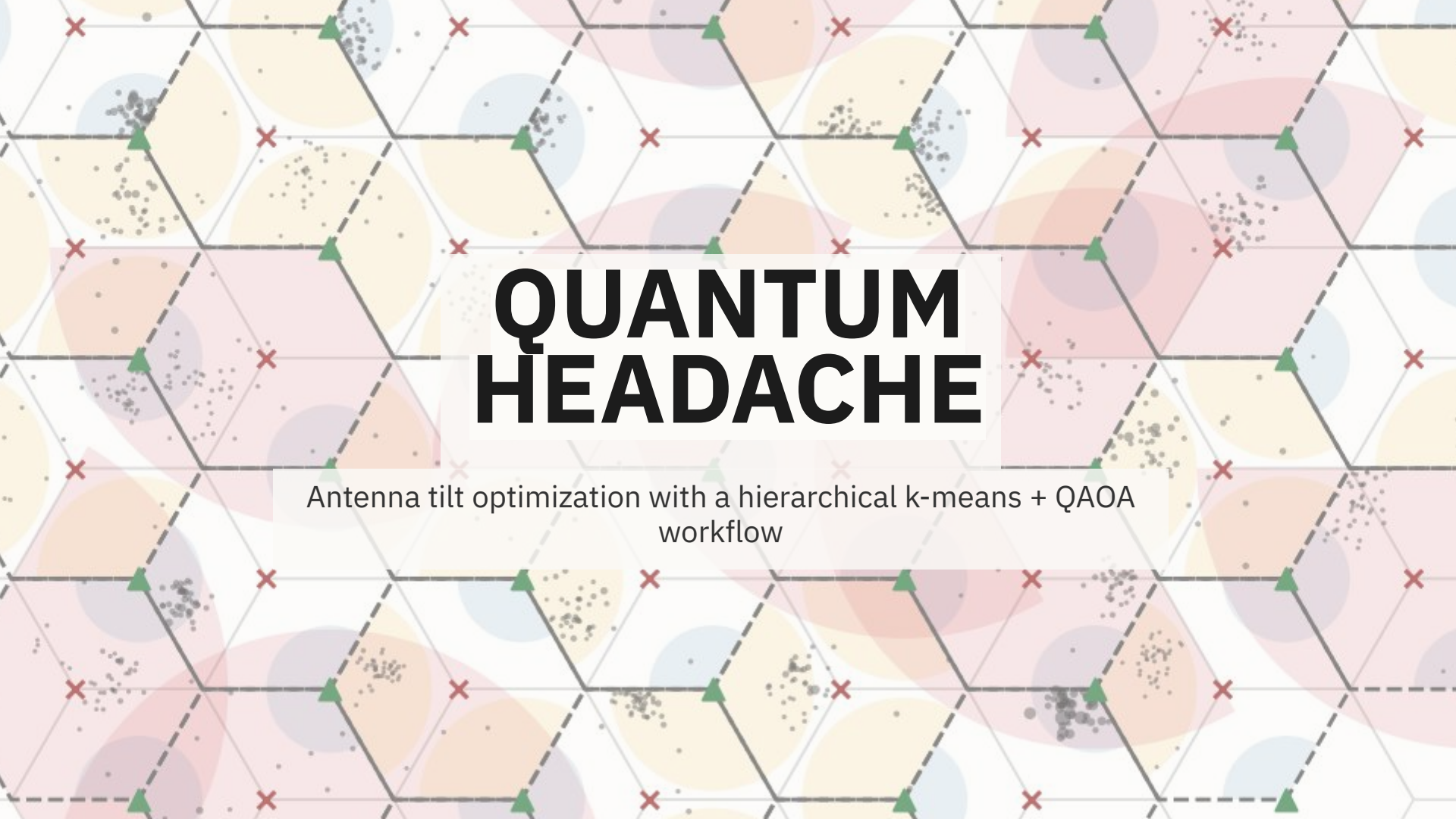
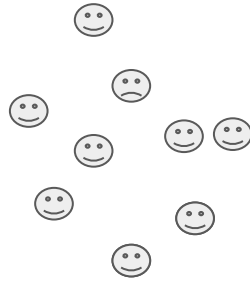




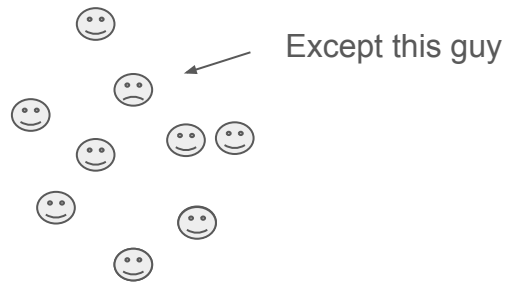
QUANTUM HEADACHE

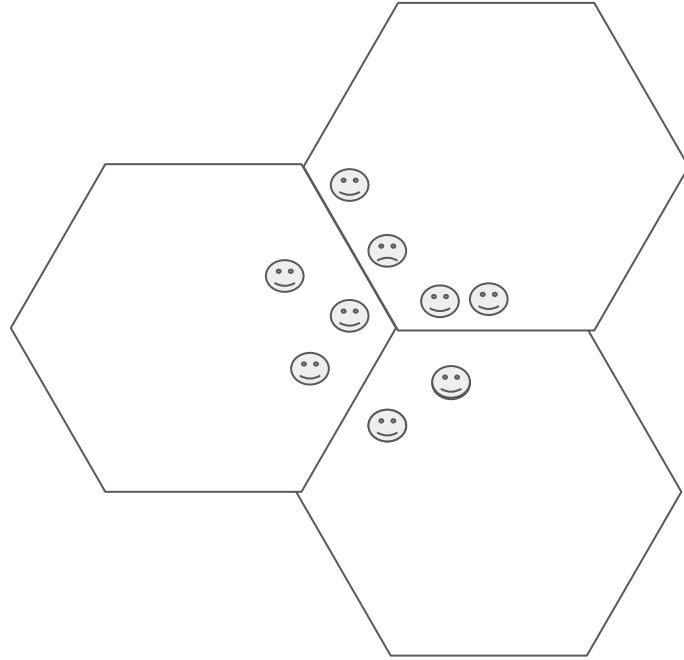
Antenna tilt optimization with a hierarchical k-means + QAOA workflow

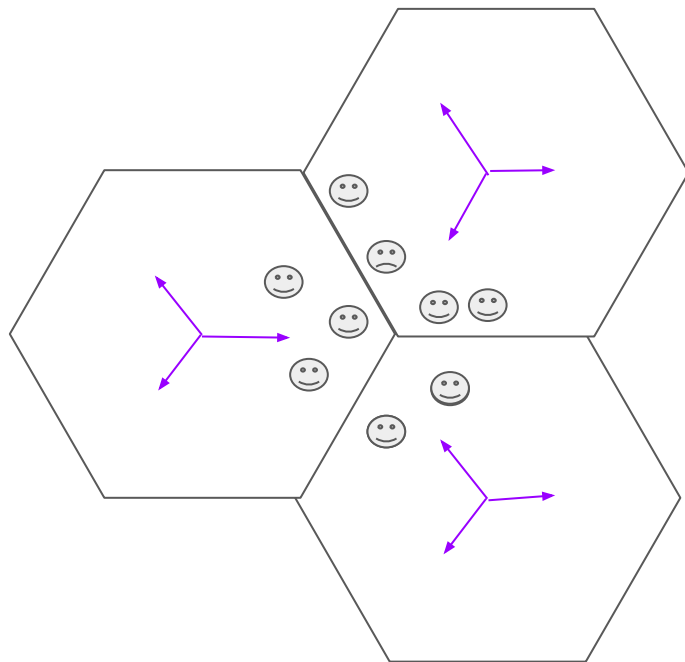
AT&T customers

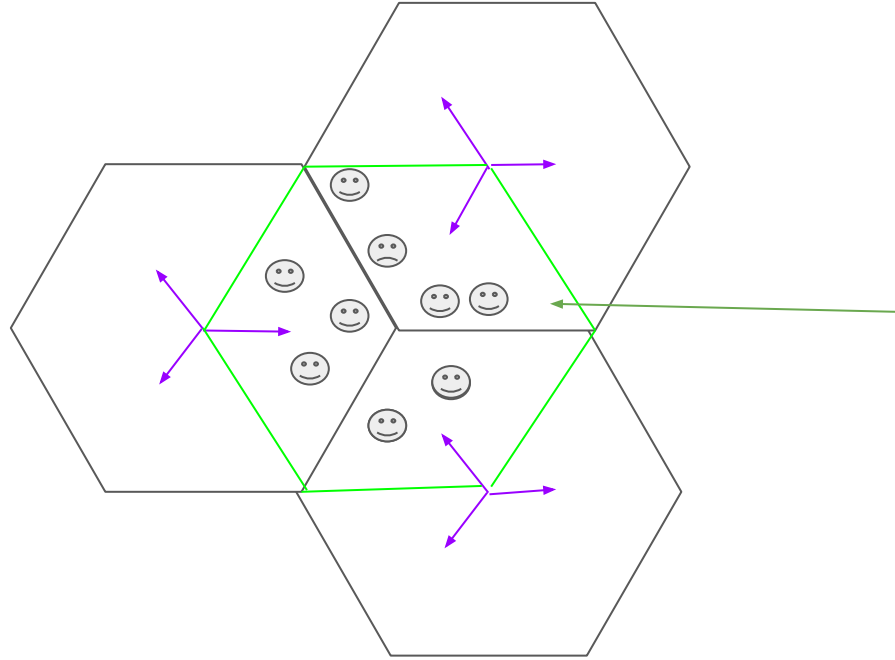


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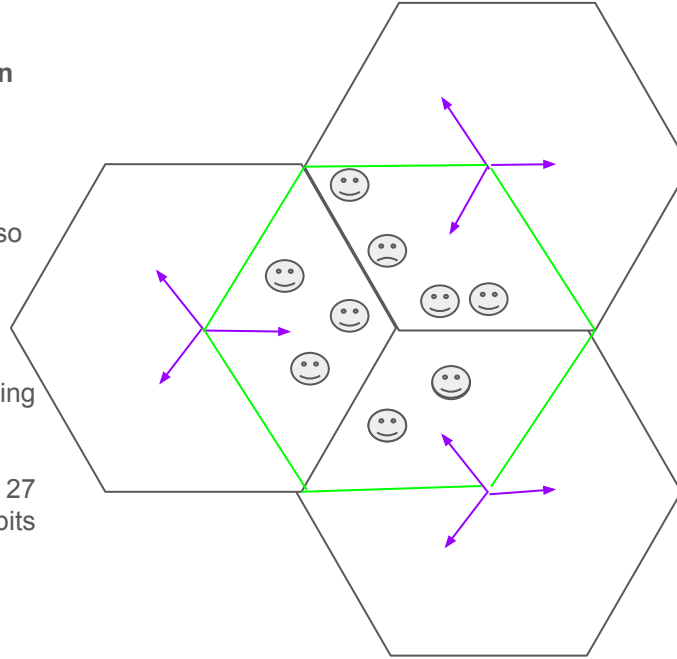




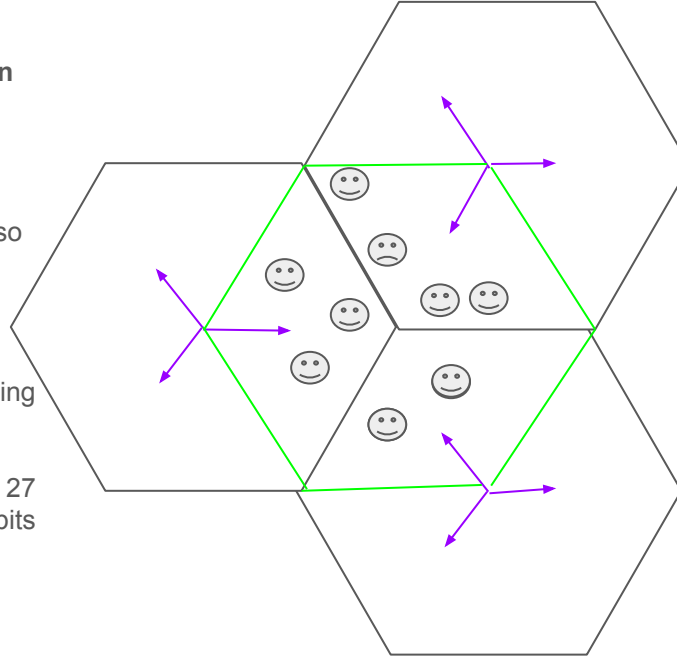


Interaction
Region ($\mathbf{R}$)

- Hexagonal cells, with 120deg cone per sector
- Each sector has 3 tilts (overall 27 config.)
- New hexagon defined as **Interaction Region (def R)**
- Objective : minimize cost function so that service quality is maximize
- This 3-orientations toy model hold most of the issues relevant - Interference, Deadzone, Overshooting and reception quality
- 3 orientation per sector per **R** gives 27 options, we encode them using 5 Qbits



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3 Demand-weighted utility

Combine reception quality, interference, and unmet demand into a single utility function.

$$U(t) = \alpha Q(t) - \beta I(t) - \gamma D(t)$$

$$Q(t) = \frac{\sum_p w_p q_p(t)}{\sum_p w_p}$$

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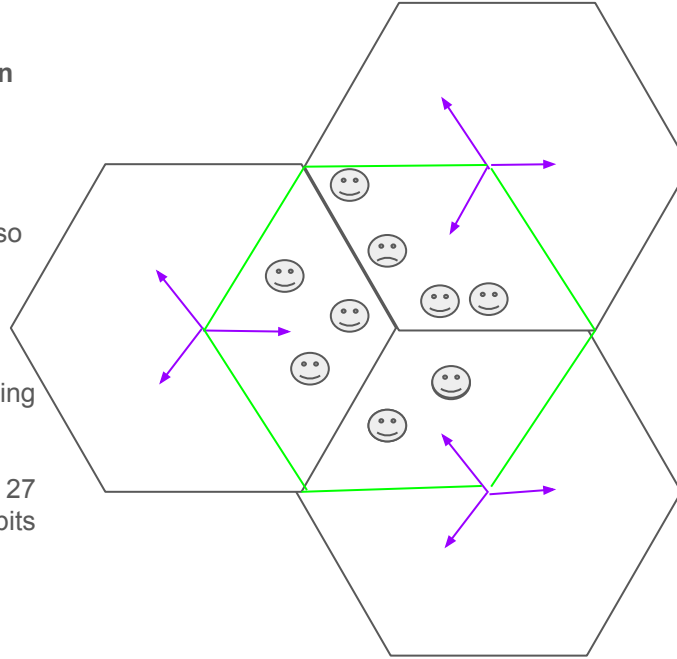
- $q_p(t)$: reception quality at point p
- $i_p(t)$: interference penalty at point p
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- w_p : demand weight at point p

Toy tilt quality scores (per sector)

Tilt	q (quality score)
Tilt 0	3
Tilt 1	2
Tilt 2	1

Dead zones matter only where demand exists ($d_p(t)$ weighted by w_p).

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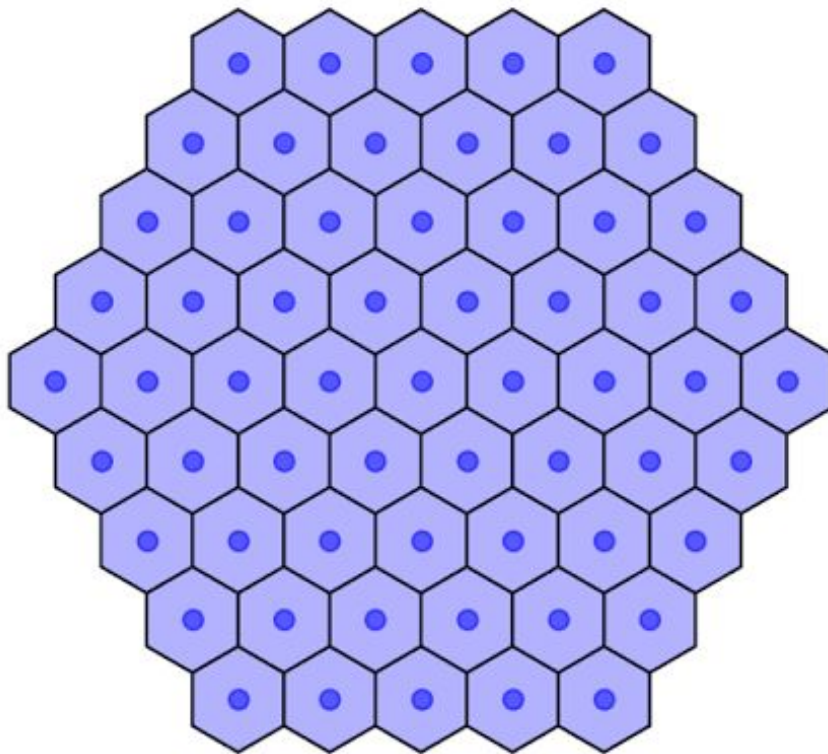
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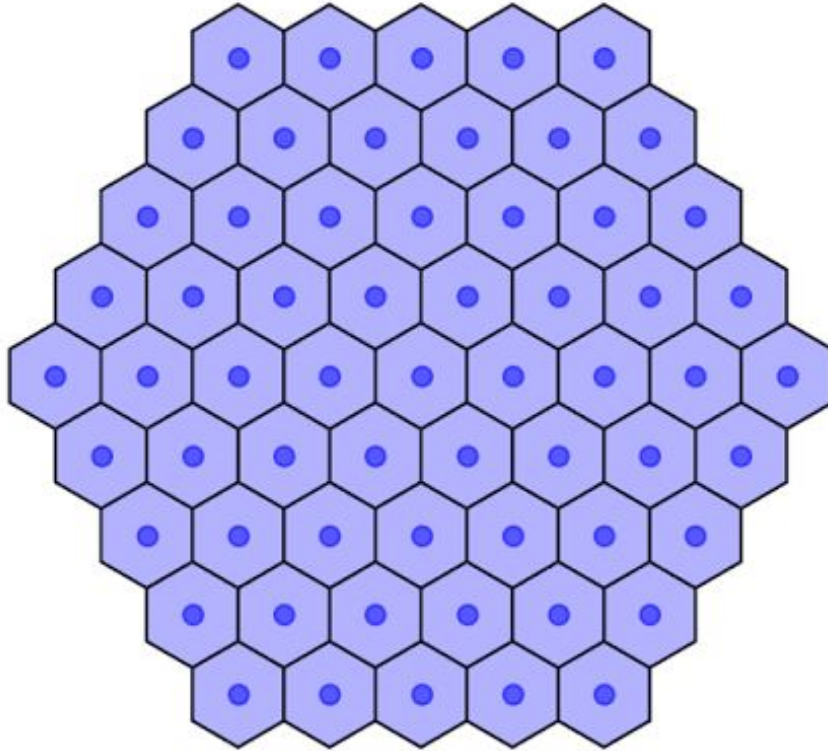
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Dead zones matter only where demand exists ($d_p(t)$ weighted by w_p).

What about the whole grid?

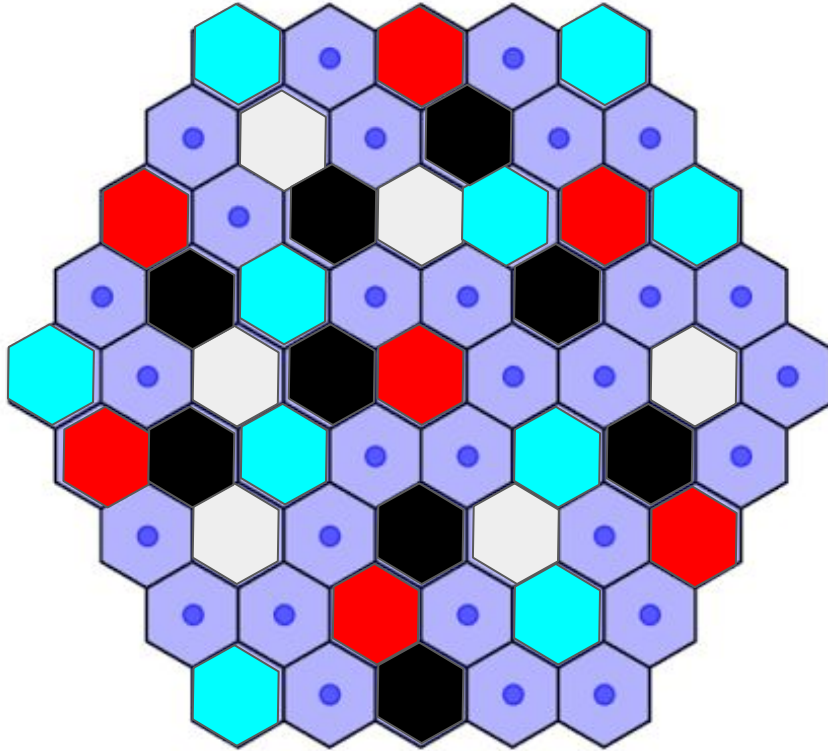


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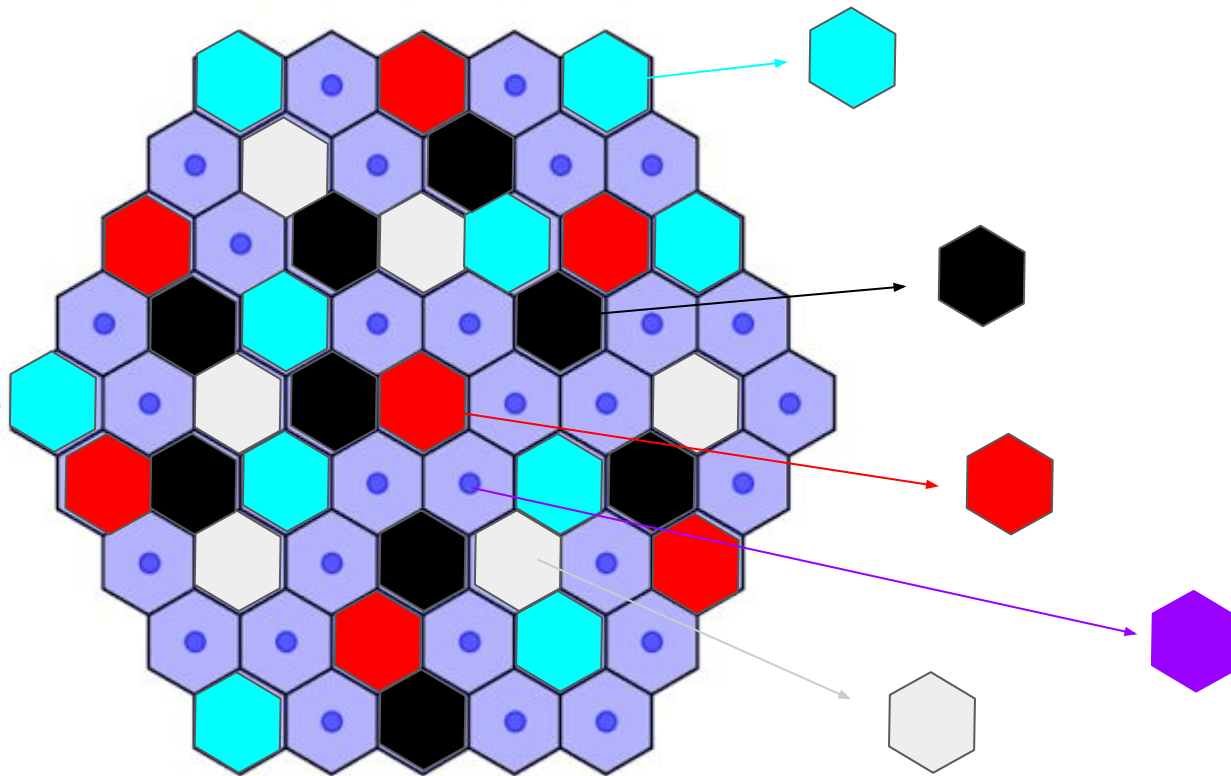
2. Use k-means to divide into 5 group (since we only have 28 QB)



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3. Sample one representative $\mathbf{R}$ from each group -> QAOA
-> apply calibration to whole group

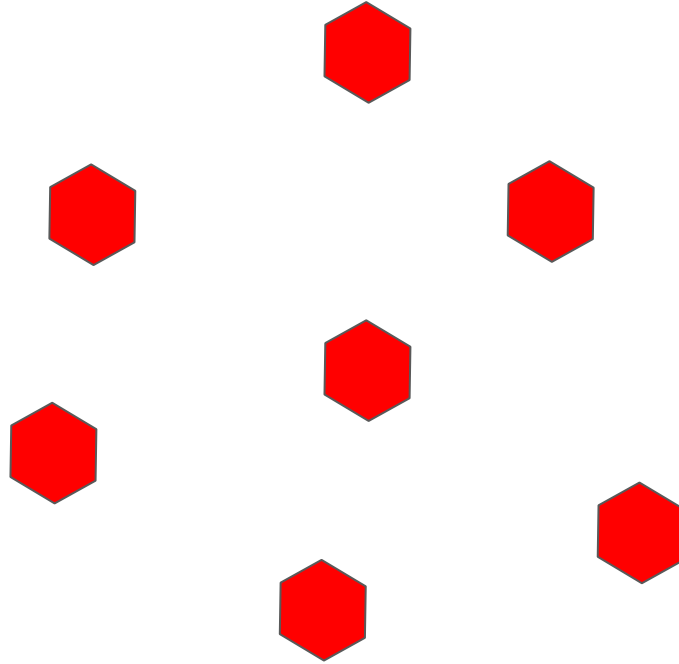


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4. Take the group with largest initial variation and start the cycle again to optimize ->

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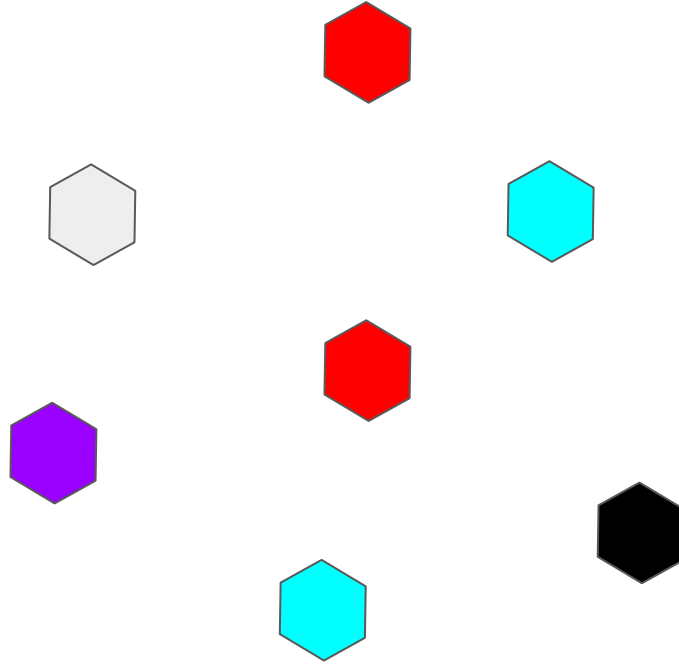
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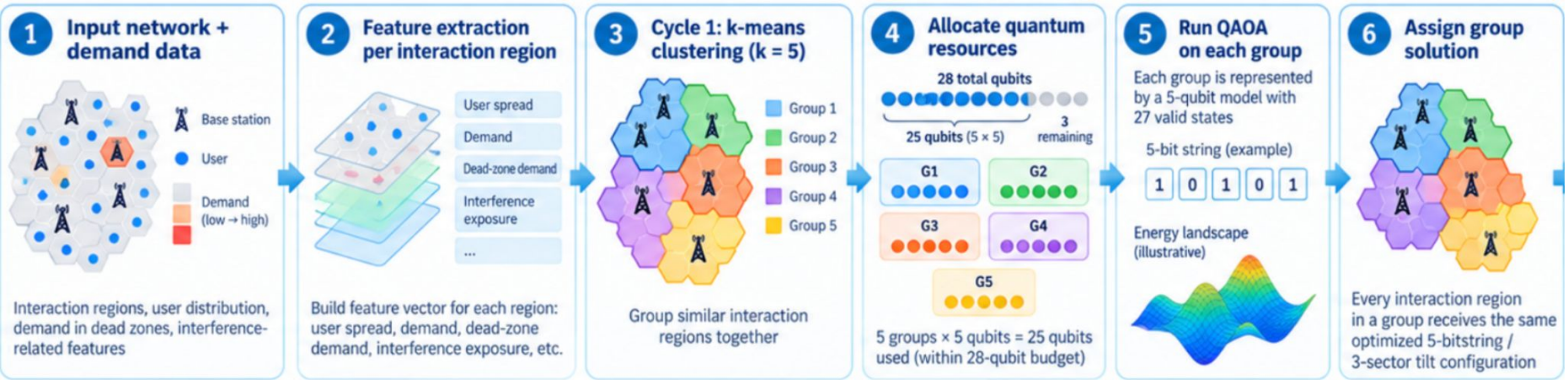
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Split with k-means -> choose repres. R -> QAOA

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2. Hybrid Optimization Flow



For group with largest initial variation

